Memo No.
Date / /

Mo Tu We Th Fr Sa Su

exo_5

Problem 1

We prove the definitions are equivalent.

(1) ⇒ (7)

If a graph is connected and has no cycle. then there cannot be two different paths between the same vertices

Example: suppose between A and B we have two paths:

A → C → B

A → D → B

Then we get a cycle:

C → B → D → A → C so this is impossible

Therefore, there is a unique path between every pair of vertices.

(7) ⇒ (1)

If there is a unique path between every pair:

graph must be connected

graph cannot have a cycle

Example: ~~the~~ if we had a cycle A → B → C → A

then between A and C we have:

Memo No.
Date / /

A → C

A → B → C

Two paths → contradiction

So graph is connected and acyclic.

(1) ⇒ (4)

In a tree. removing any edge disconnects it.

Example: A − B − C

If we remove edge B-C. C becomes isolated

So tree is minimally connected.

(4) ⇒ (1)

If removing any edge disconnects graph →

it cannot have a cycle.

Example: in a cycle: A − B − C − A

Remove A − B → graph still connected

So graph with this propety cannot contain cycles

(1) ⇒ (6)

In a tree, there is already a path between any

two vertices.

Adding a new edge creates a cycle.

Memo No.
Date / /

Example: A - B - C

Add edge A - C → cycle:

A → B → C → A so tree is maximally acyclic.

(6) ⇒ (1)

If adding any edge creates a cycle → graph must already be connected and acyclic. otherwise we could add an edge without forming a cycle.

(1) ⇒ (3)

A tree with V vertices always has V-1 edges.

Example: 3 vertices: A - B - C

Edges = 2 = 3 - 1 so it has at most V-1 edges.

(3) ⇒ (1)

If a connected graph has ≤ V-1 edges → it cannot contain a cycle.

Example: Cycle with 3 vertices: A - B - C - A

Edges = 3 > 3 - 1 so cycles need more edges.

Memo No.
Date / /

(1) ⇒ (5)

Tree has exactly V-1 edges → so ≥ V-1.

(5) ⇒ (1)

if acyclic graph has ≥ V-1 edges → it must be connected.

Examples forest ~~is~~ with 2 components:

A - B C - D

V=4, edges = 2 < 3 so to reach V-1 edges, it must be one component.

(1) ⇔ (2)

A forest is an acyclic graph.

Each connected part of a forest is a tree.

Example: forest: A - B C - D

Each part is a tree

So all definitions describe the same structure.

Memo No.
Date / /

Problem 2

Vertices: A=0, B=1, C=2, D=3, E=4.

Graph 1 (Undirected)

from CSC we get edges:

A-B
A-C
B-C
B-D
C-D
D-E

Adjacency matrix:

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

simple drawing:

A - B
\ /
C - D - E

Graph 2 (Directed)

Edges:

A → B
A → C
B → C
C → D
D → B
B → E
D → E

Adjacency matrix:

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Mo Tu We Th Fr Sa Su

Memo No.

Date / /

Unique Cycle

Look at:

$B \Rightarrow C$

$C \rightarrow D$

$D \rightarrow B$

So cycle is:

$B \rightarrow C \rightarrow D \rightarrow B$